

DSA Interview Preparation — AI / Agentic AI Engineer

50 curated Python problems for Junior/Associate AI Engineer interviews

Method: Solve first without the solution. If stuck, study the explanation, close it, and code again. Always explain your approach and Time/Space complexity.

Priority: Arrays/Hashing → Two Pointers → Sliding Window → Stack → Binary Search → Linked List → Trees → Heap → Graphs → Basic DP.

Arrays & Hashing

#	Problem	Level	Pattern	Practice	Solution
1	Two Sum	Easy	Hash Map	LeetCode	NeetCode
2	Contains Duplicate	Easy	Hash Set	LeetCode	NeetCode
3	Valid Anagram	Easy	Frequency Map	LeetCode	NeetCode
4	Group Anagrams	Medium	Hash Map	LeetCode	NeetCode
5	Top K Frequent Elements	Medium	Hash Map + Heap	LeetCode	NeetCode
6	Product of Array Except Self	Medium	Prefix/Suffix	LeetCode	NeetCode
7	Longest Consecutive Sequence	Medium	Hash Set	LeetCode	NeetCode
8	Maximum Subarray	Medium	Kadane	LeetCode	NeetCode

Two Pointers

#	Problem	Level	Pattern	Practice	Solution
9	Valid Palindrome	Easy	Two Pointers	LeetCode	NeetCode
10	Two Sum II	Medium	Two Pointers	LeetCode	NeetCode
11	3Sum	Medium	Sorting + Two Pointers	LeetCode	NeetCode
12	Container With Most Water	Medium	Two Pointers	LeetCode	NeetCode

Sliding Window

#	Problem	Level	Pattern	Practice	Solution
13	Best Time to Buy and Sell Stock	Easy	Sliding Window	LeetCode	NeetCode
14	Longest Substring Without Repeating Characters	Medium	Sliding Window + Set	LeetCode	NeetCode
15	Longest Repeating Character Replacement	Medium	Sliding Window	LeetCode	NeetCode
16	Minimum Size Subarray Sum	Medium	Sliding Window	LeetCode	NeetCode

Stack

#	Problem	Level	Pattern	Practice	Solution
17	Valid Parentheses	Easy	Stack	LeetCode	NeetCode
18	Min Stack	Medium	Stack Design	LeetCode	NeetCode
19	Daily Temperatures	Medium	Monotonic Stack	LeetCode	NeetCode

Binary Search

#	Problem	Level	Pattern	Practice	Solution
20	Binary Search	Easy	Binary Search	LeetCode	NeetCode
21	Search in Rotated Sorted Array	Medium	Binary Search	LeetCode	NeetCode
22	Find Minimum in Rotated Sorted Array	Medium	Binary Search	LeetCode	NeetCode
23	Koko Eating Bananas	Medium	Binary Search on Answer	LeetCode	NeetCode

Linked List

#	Problem	Level	Pattern	Practice	Solution
24	Reverse Linked List	Easy	Pointers	LeetCode	NeetCode
25	Merge Two Sorted Lists	Easy	Pointers	LeetCode	NeetCode
26	Linked List Cycle	Easy	Fast/Slow Pointers	LeetCode	NeetCode
27	Remove Nth Node From End	Medium	Two Pointers	LeetCode	NeetCode

Trees & BST

#	Problem	Level	Pattern	Practice	Solution
28	Maximum Depth of Binary Tree	Easy	DFS	LeetCode	NeetCode
29	Invert Binary Tree	Easy	DFS	LeetCode	NeetCode
30	Binary Tree Level Order Traversal	Medium	BFS	LeetCode	NeetCode
31	Validate Binary Search Tree	Medium	DFS / Inorder	LeetCode	NeetCode
32	Lowest Common Ancestor of a BST	Medium	BST	LeetCode	NeetCode
33	Diameter of Binary Tree	Easy	DFS	LeetCode	NeetCode

Heap / Priority Queue

#	Problem	Level	Pattern	Practice	Solution
34	Kth Largest Element in an Array	Medium	Heap	LeetCode	NeetCode
35	K Closest Points to Origin	Medium	Heap	LeetCode	NeetCode
36	Find Median from Data Stream	Hard	Two Heaps	LeetCode	NeetCode

Graphs

#	Problem	Level	Pattern	Practice	Solution
37	Number of Islands	Medium	DFS / BFS	LeetCode	NeetCode
38	Clone Graph	Medium	DFS / BFS	LeetCode	NeetCode
39	Rotting Oranges	Medium	BFS	LeetCode	NeetCode
40	Course Schedule	Medium	Topological Sort	LeetCode	NeetCode

Dynamic Programming

#	Problem	Level	Pattern	Practice	Solution
41	Climbing Stairs	Easy	1-D DP	LeetCode	NeetCode
42	House Robber	Medium	1-D DP	LeetCode	NeetCode

#	Problem	Level	Pattern	Practice	Solution
43	Coin Change	Medium	1-D DP	LeetCode	NeetCode

30-Day Suggested Order

Days	Focus	Target
Days 1–4	Arrays & Hashing	8
Days 5–6	Two Pointers	4
Days 7–9	Sliding Window + Stack	7
Days 10–12	Binary Search	4
Days 13–15	Linked List	4
Days 16–20	Trees & BST	6
Days 21–23	Heap / Priority Queue	3
Days 24–27	Graphs	4
Days 28–30	Basic DP + revision/mock interview	3 + revision

Important: This is intentionally not the full NeetCode 150. Your goal is to become safe for the DSA round while keeping your main preparation focused on AI/LLM/Agentic AI.

Solution links point to NeetCode explanations; practice links point to LeetCode.